Exam Date & Time: 15-Nov-2022 (09:00 AM - 10:00 AM)

MANIPAL ACADEMY OF HIGHER EDUCATION

B. Tech III Semester second Sessional Examination November 2022

ENGINEERING MATHEMATICS III [MAT 2155]

Marks: 15 **Duration: 60 mins.**

MCQ

Answer all the questions. Section Duration: 20 mins

1) Let $X=\{1,2,3,4\}$

. Then R={(1,1), (2, 2), (3, 3), (1,2), (2, 3), (3, 4), (4, 4), (2,4)} is a binary relation defined on X. Then R is --------

(0.5)

1)	Reflexive, symmetric and transitive		2)	Not reflexive, anti-symmetric and transitive		3)	Reflexive, not symmetric and not transitive		4)	Reflexive, anti-symmetric and transitive	

2) Let G be a self-complementary graph with 5 vertices. Then the diameter of G is…..

(0.5)

1) 2 2) 3 3) 1 4) 4

3) If a simple graph G has n vertices and m edges, then the number of edges in $\bar{G}$ is _____

(0.5)

1)	$\frac{n^2+n-2m}{2}$		2)	$\frac{n^2+2n-2m}{2}$		3)	$\frac{n^2-n-2m}{2}$		4)	$\frac{n^2-2m-2n}{2}$	

4) Let G be a simple graph with 6 vertices. The degree of 5 vertices are (2,3,3,3,5) and then number of edges of graph G_____

(0.5)

1) 7 2) 8 3) 9 4) 10

5) In a Boolean lattice, if (0.5)

$b \wedge \bar{c} = 0$

, then

1) $b \geq c$ 2) $b = c$ 3) $b \leq c$ 4) $b \leq \bar{c}$

6) Any chain with three or more elements is _____ (0.5)

1)	distributive but not complemented		2)	complemented but not distributive		3)	not a lattice		4)	Boolean lattice	

7) If a, b, and c are elements of a lattice, such that $(a \vee b) \wedge c = a$, then $a \vee c$ = _____ (0.5)

1) a 2) b 3) c 4) $b \vee c$

8) The graph in which, there is a closed trail which includes every edge of the graph is known as _____. (0.5)

1)	Hamiltonian graph		2)	Euler graph		3)	Planar graph		4)	Complete graph	

9) Let S_{24} be the set of all positive divisor of 24 with Poset $(S_{24}, /)$. Then the set of atoms are (0.5)

1) 2,6 2) 2,3 3) 2 4) 8,12

10) If a tree has 7 edges, then the no. of vertices in that tree is…….. (0.5)

1) 8 2) 7 3) 6 4) 4

DES

Answer all the questions.

21) If G is a graph, then show that either G or $\bar{G}$ is connected. (2)

22) In a distributive lattice, if an element has a complement , then it is unique. (2)

23) Let $E(x_1, x_2, x_3) = (x_1 \wedge x_2) \vee (x_1 \wedge x_3) \vee (\overline{x_2} \wedge x_3)$ be a Boolean expression over $(\{0,1\}, \wedge, \vee, -)$. Write the Boolean expression in both DNF and CNF . (3)

24) Show that a connected graph is bipartite if and only if it has no odd cycles. (3)

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